ABSTRACTOR

A Dissertation submitted in partial fulfillment of the requirement for the award of the degree of

MASTER OF COMPUTER APPLICATIONS

Of

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

By

Mr. SYED MAFAAZ

4SN24MC037

Under the Guidance of

Dr. DHEERAJ.H

DEPARTMENT OF MASTER OF COMPUTER APPLICATIONS

SRINIVAS INSTITUTE OF TECHNOLOGY

(Affiliated to VTU, Belagavi, Approved by AICTE, New Delhi,

Recognized by Govt. of Karnataka)

Mangalore -574143, Karnataka

DECEMBER 2025

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DEPARTMENT OF MASTER OF COMPUTER APPLICATIONS

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CERTIFICATE

This is to certify that, ***Mr. SYED MAFAAZ*** *bearing* ***USN 4SN24MC037*** *has successfully completed his third semester project work entitled "****ABSTRACTOR "*** *as a partial fulfillment for the award for Master of Computer Applications degree, during the academic year 2025-2026.*

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1.

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DECLARATION

I, **Mr. SYED MAFAAZ**, student of 3rd Semester MCA, **Srinivas Institute of Technology, Valachil**, bearing USN: **4SN24MC037**, hereby declare that I have carried out the project entitled "**ABSTRACTOR**" under the supervision of **Dr. Dheeraj H**, Associate Professor, Department of Master of Computer Applications, and submitted the same in partial fulfillment of the requirements for the award of the degree of Master of Computer Applications during the academic year 2025–2026, from Visvesvaraya Technological University, Jnana Sangama, Belagavi.

This work which is being presented in the project report submitted to the Department of MCA from Srinivas Institute of Technology, Valachil, Mangalore, is an authentic record of project work.

Place: Mangalore
Date:

Mr. Syed Mafaaz
Signature:

ACKNOWLEDGEMENT

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ABSTRACT

In the present digital era, the exponential growth of textual data across articles, reports, research papers, and online content has created a critical need for efficient information condensation techniques. Manual summarization is time-consuming, error-prone, and impractical for large volumes of text. To address this challenge, this project presents ABSTRACTOR – An AI-Powered Text Summarization Tool, a web-based system designed to automatically generate concise, meaningful summaries from large textual inputs using modern Natural Language Processing (NLP) techniques.

The proposed system supports both extractive and abstractive summarization approaches, with a primary focus on deep learning–based abstractive summarization using the BART (Bidirectional and Auto-Regressive Transformer) model. The system is capable of processing plain text as well as document files such as PDF, DOCX, and TXT, making it suitable for real-world academic and professional use cases. A robust text chunking algorithm is implemented to handle large documents by splitting them into manageable segments while preserving contextual continuity.

The architecture follows a three-tier design, consisting of a React-based frontend for user interaction, a Flask backend for request handling and processing, and a service layer responsible for text extraction, preprocessing, summarization, and fallback mechanisms. When external AI services are unavailable or fail to respond, the system seamlessly switches to a TF-IDF-based extractive summarization method, ensuring reliability and uninterrupted service. Additionally, the application maintains summary history using browser storage, enhancing usability and user experience.

Overall, ABSTRACTOR demonstrates how transformer-based language models can be effectively integrated into a scalable web application to deliver fast, accurate, and user-friendly text summarization. The system significantly reduces reading time while preserving essential information, making it a valuable tool for students, researchers, and professionals dealing with large textual datase